PolicyClaw — System Analysis Documentation (SAD)

Project	PolicyClaw — AI insurance decision copilot for Malaysians
Event	UMHackathon 2026 — Domain 2 (AI for Economic Empowerment & Decision Intelligence)
Document version	1.0 — 2026-04-26
Status	Active — Hackathon MVP submission
Paired specs	`PRD.md` v2.2, `QATD.md` v1.0
Tagline	*Claw through complexity. Decide with confidence.*

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1. Introduction

PolicyClaw is an AI insurance decision copilot built for the 2024–2026 Malaysian repricing crisis. A user uploads 1–3 policy PDFs, confirms auto-detected profile fields, and receives a verdict — **Hold / Switch / Downgrade / Add Rider** — grounded in four sequential `gpt-5-mini` calls (Extract, Annotate, Score, Recommend), every output cited to a specific clause and page.

1.1 Purpose

This System Analysis Documentation describes the technical scope and design decisions behind PolicyClaw's 24-hour hackathon MVP. It is the engineering counterpart to the product spec (`PRD.md`) and the quality plan (`QATD.md`). Readers should treat the PRD as authoritative for *what* must ship and *why*; this document is authoritative for *how* it ships.

The key system elements covered:

1. **Architecture** — Two-tier web application (Next.js 15 frontend ↔ FastAPI backend) deployed on a single demo machine. The reasoning layer is a streaming OpenAI `gpt-5-mini` integration; persistence is ephemeral cache + browser `localStorage` (Supabase is the post-hackathon ship target).
2. **Data Flows** — How a PDF moves from the browser uploader through PyMuPDF parsing, four staged LLM calls (each cache-keyed by SHA-256), Monte Carlo simulation in numpy, and finally back to the browser as an `AnalyzeResponse` plus a deferred ClawView annotation payload.
3. **Model Process** — End-to-end LLM orchestration: prompt construction, context-window assembly from `PolicyChunk` lists, SSE-streamed responses, Pydantic validation, and a deterministic mock/heuristic fallback when the upstream model is unreachable.
4. **Role of Reference** — A single source of truth for the implementing developer, the QA reviewer, and the hackathon judge to align on architectural intent without reading the entire codebase.

1.2 Background

Malaysia's medical insurance market entered a repricing crisis in 2024: 9% of MHIT policyholders saw premium hikes greater than 40% (BNM 2024 Annual Report); cumulative claims inflation reached 56% across 2021–2023; and Bank Negara Malaysia's interim 10% annual cap window expires **31 December 2026**. Most policyholders do not know they qualify for the cap. Combined with the 30 million Malaysians EY estimated as underinsured in 2024, the root problem is decision paralysis, not affordability — there is no neutral, explainable second opinion in Bahasa Malaysia or English. PolicyClaw addresses that gap.

Previous version. None — this is the first build, scaffolded from scratch during UMHackathon 2026.

Changes in major architectural components. The reasoning provider was swapped during the build. The project initially targeted **Z.AI GLM via Ilmu (`api.ilmu.ai` / `ilmu-glm-5.1`)** to satisfy the hackathon's then-mandatory Z.AI rule. The Ilmu gateway proved unstable in practice; the organizers waived the Z.AI requirement for this submission, and the reasoning layer was re-pointed to **OpenAI `gpt-5-mini`** on `https://api.openai.com/v1`. The shared `glm_client.py` retains the `post_glm_with_retry` name (single-line transport contract) but routes to OpenAI; payload adaptation strips the `temperature` / `top_p` keys that `gpt-5-*` reasoning models reject and injects `reasoning_effort: "low"`.

New capabilities introduced in this build.

- **ClawView (Wow Factor 1)** — color-coded green/yellow/red risk highlights overlaid on the original PDF using PyMuPDF bounding boxes plus an LLM annotation pass.
- **FutureClaw (Wow Factor 2)** — an interactive 10-year Monte Carlo simulator with two toggleable modes (Affordability and Life Event), 1000-run numpy simulation, and an LLM narrative pass that runs once per scenario change rather than once per slider tick.
- **Demo-cache reliability layer** — every LLM stage is cache-keyed by SHA-256 of its canonical input so that identical reruns return byte-identical outputs, protecting both verdict consistency (PRD F7) and live-demo wifi drops.

1.3 Target Stakeholders

Stakeholder	Role	Expectations
Policyholder (primary user)	Aisyah, 38, M40 KL marketing manager — 2–4 active policies, just received a repricing notice with 30 days to decide.	Plain-language verdict, citation on every claim, BM/EN toggle, downloadable Action Summary, no commission-driven nudges.
Future advisor (secondary user)	Licensed agent or family member assisting a policyholder.	Confidence-scored output that flags low-certainty cases for human follow-up; no auto-purchase or auto-cancel.
Insurance regulator (judging proxy)	BNM-style oversight perspective applied by hackathon judges.	"Decision support, not advice" disclaimer on every recommendation screen; PDPA-compatible privacy posture; no PII in logs.
Development team (solo dev)	Single developer building the 24-hour MVP.	Modular FastAPI services, typed Pydantic contracts, single LLM entry point (`glm_client.py`), deterministic fallbacks for offline iteration.
QA reviewer	The author of `QATD.md` validating risk + test posture.	Defined endpoint contracts, schema-validated responses, pytest coverage on extraction / simulation / verdict paths, CI gating on smoke import + frontend build.
Hackathon judges	UMHackathon 2026 panel (Domain 2).	Architecture aligned with the SAD template, evidence of LLM centrality, working ClawView + FutureClaw demos, defensible technology choices.

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2. System Architecture & Design

2.1 High Level Architecture

2.1.1 Overview

Type	Details
System	Web Application (responsive, desktop-first; tested down to iPhone 14 viewport)
Architecture	Two-tier client–server with a streaming AI service layer. Single-machine demo; no microservice split.

PolicyClaw is structured as a client–server web application. There is one client surface — a Next.js 15 App Router single-page experience served from `localhost:3000` — and one backend — a FastAPI service on `localhost:8000`. The two communicate over JSON / multipart HTTPS. The backend wraps three internal subsystems: a PDF ingestion pipeline (PyMuPDF), a four-call LLM orchestrator (OpenAI `gpt-5-mini` via streaming SSE), and a pure-Python Monte Carlo simulator (numpy + scipy). Persistence in the MVP is **ephemeral**: process memory plus a SHA-256-keyed JSON cache under `backend/data/demo_cache/`.

Browser state lives in React + Zustand + `localStorage`. Supabase (Postgres + pgvector + Auth + Storage + Realtime) is the post-hackathon ship target — explicitly **not MVP-gating** per PRD §8.2.

```
┌──────────────────────────────────────────┐
│ User (Malaysian policyholder)            │
│ Uploads 1–3 PDFs, fills profile, reviews │
└──────────────────────────────────────────┘
                   │
                   │ HTTPS / multipart
                   ▼
┌──────────────────────────────────────────┐
│ Next.js 15 — localhost:3000              │
│ App Router · React 19 · TypeScript 5.8   │
│ Tailwind · Recharts · Zustand · Framer   │
│ Motion · react-pdf-viewer · jsPDF        │
└──────────────────────────────────────────┘
                   │
                   │ fetch JSON
                   ▼
┌──────────────────────────────────────────┐
│ FastAPI — localhost:8000                 │
│ /api/extract-policy-profile              │
│ /api/analyze   (3-call backend pipeline) │
│ /v1/clawview   (4th LLM call — Annotate) │
│ /v1/simulate/affordability               │
│ /v1/simulate/life-event                  │
└──────────────────────────────────────────┘
   │                 │                    │
   ▼                 ▼                    ▼
┌──────────┐ ┌────────────────────┐ ┌────────────────┐
│ PyMuPDF  │ │ OpenAI             │ │ numpy + scipy  │
│ parser   │ │ gpt-5-mini         │ │ Monte Carlo    │
│ (chunks  │ │ streamed via httpx │ │ (simulation)   │
│ + bbox)  │ │ + retry/fallback   │ │                │
└──────────┘ └────────────────────┘ └────────────────┘
                      │
                      ▼
              ┌────────────────┐
              │ Local demo cache │
              │ backend/data/    │
              │ demo_cache/*.json│
              └────────────────┘
```

2.1.2 LLM as a Service Layer

The reasoning layer is **not** a generic "AI module" hung off the side of the architecture. `gpt-5-mini` plays four distinct roles in the request lifecycle, each with its own prompt, its own Pydantic response schema, and its own deterministic fallback. Removing the LLM removes plain-language extraction, ClawView clause annotations, Health Score sub-scores, and the recommendation narrative. Only the Monte Carlo numbers survive — and even those lose their interpretation. This is the LLM-centrality test the hackathon brief enforces.

LLM call	Endpoint that fires it	Input	Output	Failure mode
1. Extract	`POST /api/analyze` (stage 1)	Raw PDF text + user profile	Structured `Policy` Pydantic model — insurer, premium, coverage, riders, exclusions, waiting periods	`_mock_policy_xray` returns LOW-confidence canned shape
2. Annotate	`POST /v1/clawview` (frontend-triggered after Analyze returns)	Bounding-boxed clauses	List of `ClawViewAnnotation` with `risk_level ∈ {green, yellow, red}` + plain explanation + clause id	Heuristic risk classifier seeded by keyword rules
3. Score	`POST /api/analyze` (stage 2)	Extracted policy + profile	`HealthScore` with 4 sub-scores (Coverage Adequacy, Affordability, Premium Stability, Clarity & Trust)	`_heuristic_health_score` calibrated to land in the 30–85 band per PRD F5
4. Recommend	`POST /api/analyze` (stage 3)	Extracted policy + score + 10-year projected cost	Verdict + 3 cited Reasons + confidence band + MYR impact + `needs_rider` flag	Deterministic `generate_verdict` heuristic with LOW confidence band

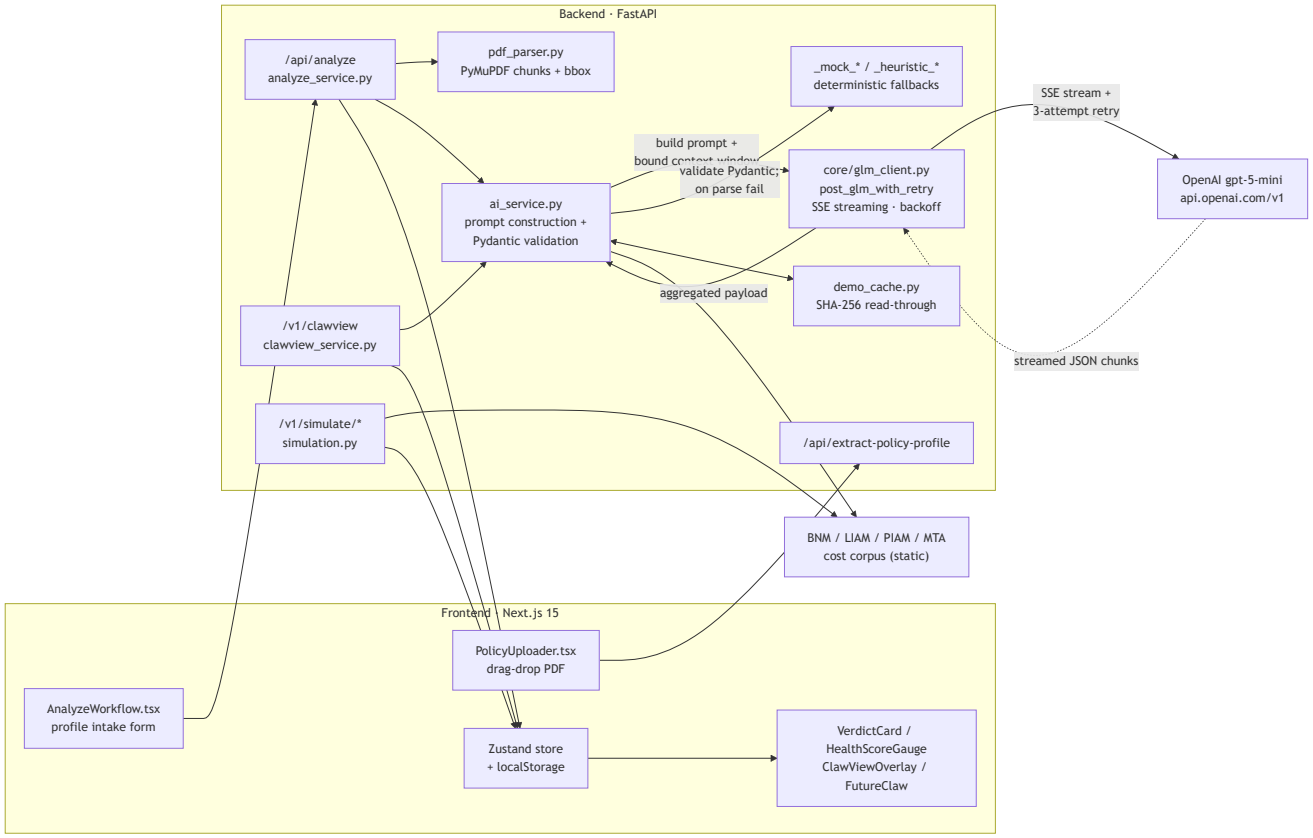
A separate single-batch LLM call powers FutureClaw life-event narratives (`futureclaw_narrative.py`) — bilingual (BM + EN) one-shot generation of the four scenario narratives, fired only on user mode toggle, not on slider drag. This is intentionally *outside* the four-call pipeline because it is interactive and triggered by the user, not by analysis.

Token / context limits and chunking. PyMuPDF chunks each PDF into `PolicyChunk` objects with page + section + bbox. The Extract call sees the full chunk list (typical demo policy: ~40 chunks, ~12k tokens). For longer policies, `rag.py` performs lexical retrieval to keep the context window under the model's effective limit. The Annotate call is per-chunk-batch and is tightened to **2 attempts / 30s** (vs. the default 3 attempts / 120s) so a slow upstream response degrades to the heuristic mock within ~60s instead of stalling the frontend.

Demo cache as part of the LLM contract. Each of Extract, Score, Recommend, and Annotate is read-through to `backend/data/demo_cache/<stage>_<sha256>.json` — identical inputs produce identical outputs across fresh processes. This is what gives PRD F7 verdict consistency without relying on `temperature` (which the `gpt-5-*` reasoning model rejects).

2.1.3 Dependency Diagram

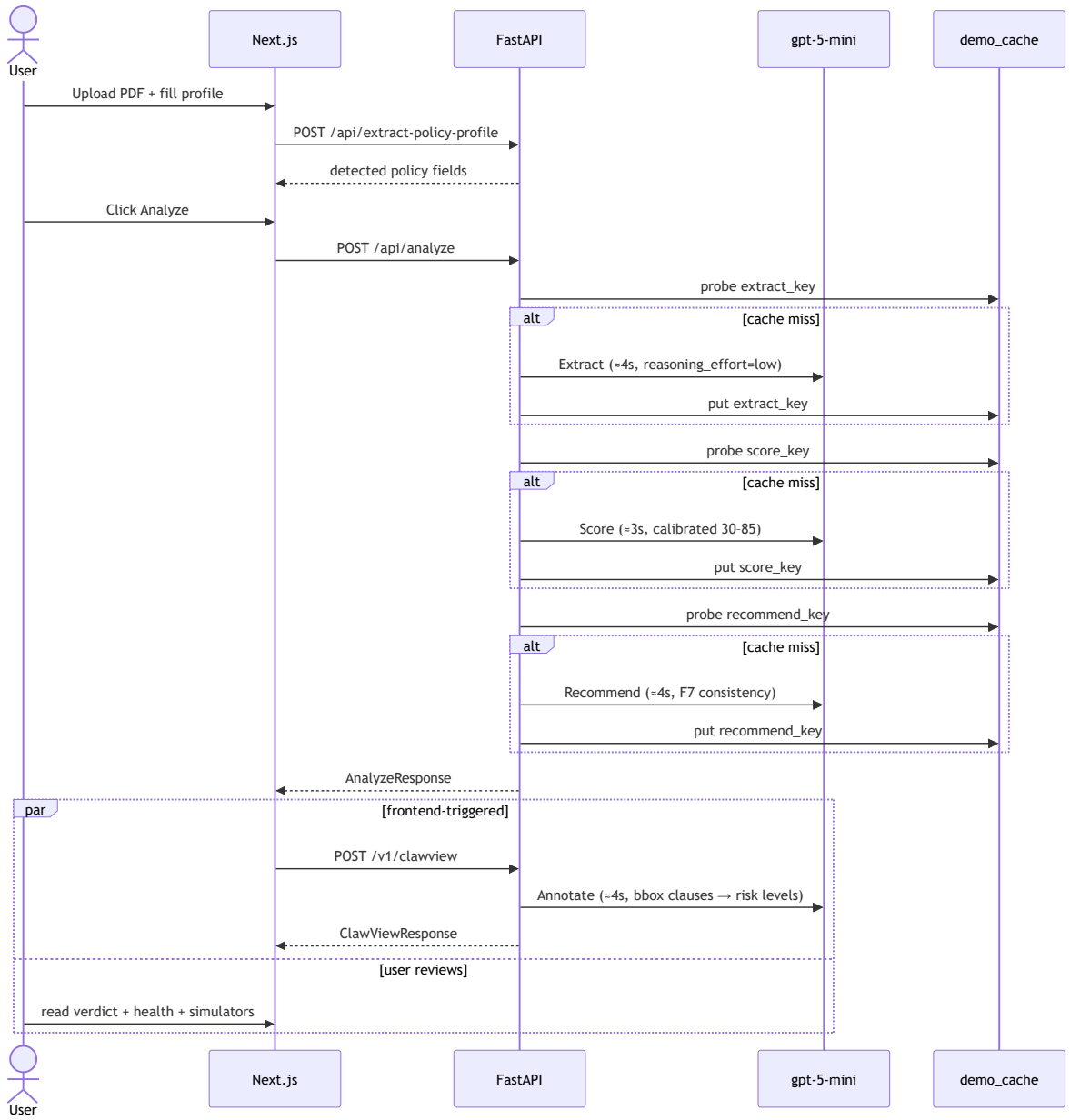
How prompts are constructed, sent, parsed, and routed back through the system. Distributed components are not in scope (single-machine demo); the diagram emphasizes the AI service layer.



Reading guide. The user uploads via `PolicyUploader.tsx`, which calls `/api/extract-policy-profile` to auto-fill the intake form. On *Analyze*, `/api/analyze` orchestrates the three sequential LLM calls; each call funnels through `ai_service.py` for prompt construction, then through `glm_client.post_glm_with_retry` for streaming transport, then back through `ai_service` for Pydantic validation. The cache layer wraps every stage. ClawView (`/v1/clawview`) is fired separately by the frontend in parallel with the user reviewing the verdict — the fourth LLM call does not block perceived end-to-end latency. The Monte Carlo simulator never invokes the LLM during a slider drag; it only triggers a narrative regeneration on mode toggle.

2.1.4 Sequence Diagram — Analyze Flow

Walks through the `Aisyah uploads a medical policy` user journey from intake to verdict. Demonstrates the 3+1 LLM call orchestration and the parallel ClawView fetch.


User
Next.js
FastAPI
gpt-5-mini
demo_cache
Upload PDF + fill profile
POST /api/extract-policy-profile
detected policy fields
Click Analyze
POST /api/analyze
probe extract_key
alt
[cache miss]
Extract (≈4s, reasoning_effort=low)
put extract_key
probe score_key
alt
[cache miss]
Score (≈3s, calibrated 30-85)
put score_key
probe recommend_key
alt
[cache miss]
Recommend (≈4s, F7 consistency)
put recommend_key
AnalyzeResponse
par
[frontend-triggered]
POST /v1/clawview
Annotate (≈4s, bbox clauses → risk levels)
ClawViewResponse
[user reviews]
read verdict + health + simulators
User
Next.js
FastAPI
gpt-5-mini
demo_cache

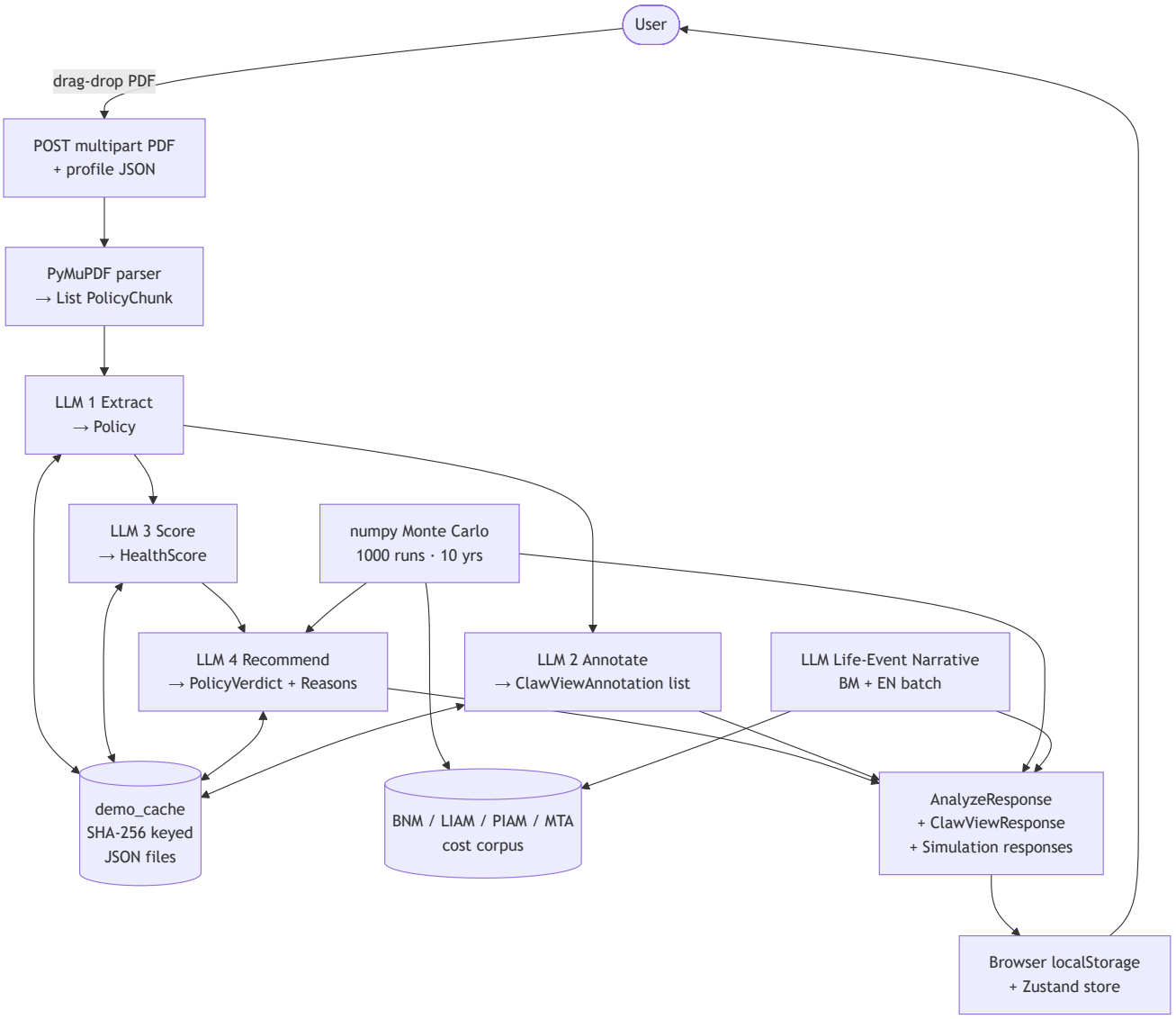
2.2 Technological Stack

Layer	Choice	Rationale
Frontend	Next.js 15 (App Router), React 19, TypeScript 5.8, Tailwind, shadcn/ui, react-pdf-viewer, Recharts, Zustand, TanStack Query, Framer Motion, jsPDF	App Router gives us route-level code splitting; react-pdf-viewer renders the policy PDF inline so the ClawView SVG overlay can be portaled into it; Recharts handles the FutureClaw line + bar charts; Zustand keeps the BM/EN language toggle coherent without reducer boilerplate; jsPDF generates the one-page Action Summary client-side in <2s with no backend round-trip.
Backend	Python 3.12, FastAPI, Pydantic v2, httpx (streaming), numpy, scipy, PyMuPDF (fitz)	Typed async endpoints with auto-generated OpenAPI at `/docs`; Pydantic v2 models double as the LLM response contracts; PyMuPDF returns per-clause bounding-box coordinates that ClawView's SVG overlay needs — pypdf does not.
AI	OpenAI `gpt-5-mini` on `https://api.openai.com/v1` via streaming SSE through `core/glm_client.py:post_glm_with_retry`. 3 attempts × 120s read timeout default; Annotate tightened to 2 × 30s.	Streaming SSE keeps long reasoning-model responses healthy. Reasoning models reject `temperature` / `top_p` — the shared client strips them and injects `reasoning_effort: "low"` automatically (see `_adapt_payload_for_provider` in `glm_client.py`).
Database (MVP)	None — process memory + browser `localStorage` + SHA-256-keyed JSON cache under `backend/data/demo_cache/`	A 24-hour build cannot afford schema-migration risk. The cache layer guarantees both verdict consistency and offline-safe demos.
Database (ship target)	Supabase: managed Postgres + pgvector + Auth (magic-link) + Storage (private bucket with signed URLs) + Realtime + Row Level Security	Unlocks RAG over the BNM / LIAM / PIAM / MTA corpus, PDPA-compatible deletion endpoint, and live-stream agent results during a future demo. Explicitly **not MVP-gating** per PRD §8.2.
Cloud / Deployment	Local development on the demo laptop. CI on GitHub Actions (`ubuntu-latest`, Python 3.12 + Node 20).	Frontend deployment (Vercel) and backend deployment (Render / Fly) are post-hackathon. Localhost demo eliminates network-fault surface during judging.

2.3 Key Data Flows

2.3.1 Data Flow Diagram (DFD)

How data moves across the system and how it is structured at each hop.

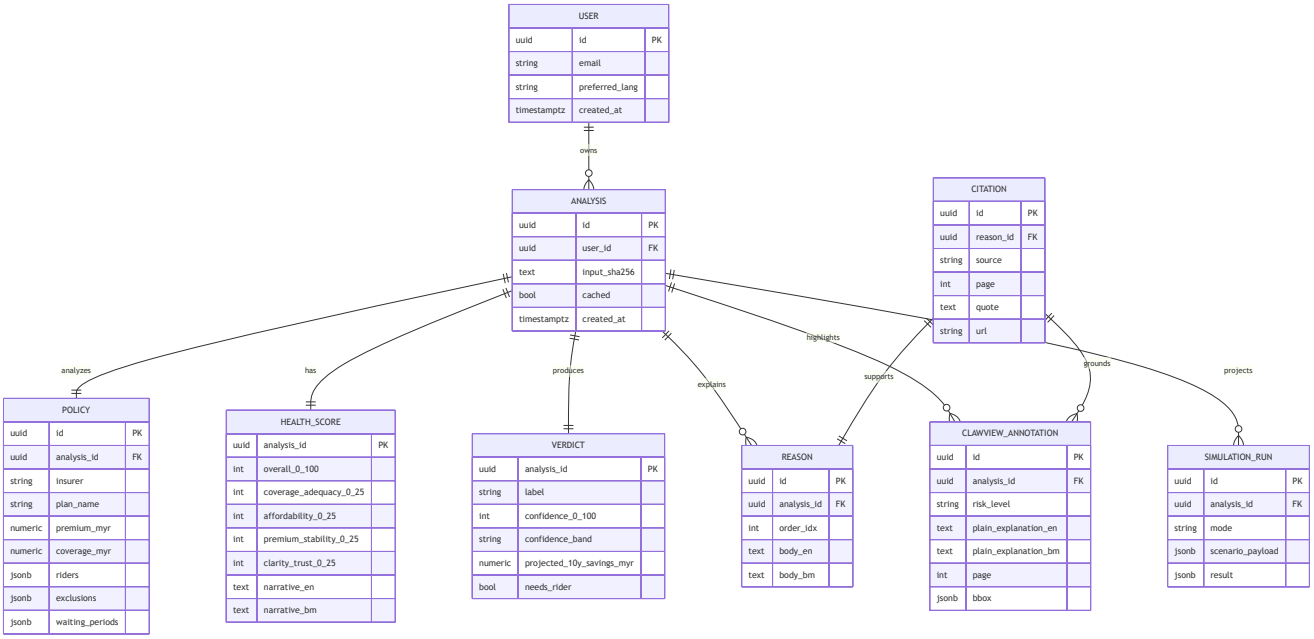


Black-hole / sink check. Every store has both an in- and out-edge. `demo_cache` is bi-directional (read on probe, write on miss). `localStorage` reads on session resume, writes on each step. The static corpus is read-only by design (curated pre-hackathon).

No-PII guarantee. Logs use UUID `analysis_id` only. Uploaded PDFs are held in process memory only for the lifetime of the analysis call, then dropped. The PDPA-compatible deletion endpoint is part of the Supabase ship target, not the MVP.

2.3.2 Normalized Database Schema (ERD)

The MVP has no database — frontend state + backend cache only. The ERD below describes the **post-hackathon ship target** in 3NF, against which the in-memory contracts already align so the migration is mechanical when Supabase is enabled.



3NF notes. `REASON` is a junction-style child of `ANALYSIS` rather than nesting in `VERDICT`, so reasons can be reordered or paginated independently. `CITATION` is referenced rather than embedded — the same clause can ground both a Reason in the verdict and a ClawView annotation, eliminating duplication. `SIMULATION_RUN.result` is JSONB rather than a wide column set because the affordability and life-event modes have different shapes (sliders vs. scenario picker); promoting to typed tables is post-MVP. Static cost data (BNM / LIAM / PIAM / MTA) is intentionally **not** modeled here — it lives as version-pinned JSON in `backend/data/bnm_corpus/` per PRD P5 (Malaysian-first, real cited data).

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3. Functional Requirements & Scope

3.1 Minimum Viable Product

The 24-hour build commits to five demonstrable features, each mapped to a PRD F-number for traceability. Every feature is P0 (ship-or-die) except multilingual which is P1.

#	Feature	Description
1	**ClawView (Wow Factor 1, F4)**	Color-coded green / yellow / red risk highlights overlaid on the original PDF using PyMuPDF bounding boxes plus the LLM Annotate call. Click a highlight → tooltip with plain-language explanation + citation. Acceptance: ≥8 highlights per demo policy, color coding accurate (red ≠ random), no offset bugs.
2	**FutureClaw (Wow Factor 2, F6)**	Interactive 10-year Monte Carlo simulator with two toggleable modes. *Affordability* — premium vs. income trajectory under inflation sliders, flags year premium > 10% of income. *Life Event* — Cancer / Heart Attack / Disability / Death scenarios with covered / co-pay / out-of-pocket bars and bilingual narrative. 1000 numpy runs, no LLM in the slider loop.
3	**Policy Health Score (F5)**	0–100 gauge with four 0–25 sub-scores (Coverage Adequacy, Affordability, Premium Stability, Clarity & Trust). Calibrated to land sample policies in the 30–85 band — not all 90+ — so the score has signal.
4	**Verdict + Citations (F7)**	One of *HOLD / DOWNGRADE / SWITCH / ADD RIDER* with three cited reasons, a 0–100% confidence score, projected 10-year MYR impact, and a "Not financial advice" disclaimer. Verdict is byte-identical across three reruns of the same input (cache-enforced determinism).
5	**Action Summary download (F8)**	Client-side jsPDF generation of a one-page summary (verdict, top 3 reasons, top 3 actions, disclaimer, `analysis_id`) in <2s with no backend round-trip.

Multilingual support (F9 — BM + EN, JSON-dict i18n, no `i18next`) is P1 but already shipped: backend strings are returned in both languages (`narrative_en`, `narrative_bm`); frontend toggles via a 40-line `t(key, lang)` helper. Polish (F10 — Framer Motion transitions) is P0 and shipped on key screens.

3.2 Non-Functional Requirements

Quality	Requirement	Implementation
Scalability	Single-machine demo handles a sustained sequence of analyses with one concurrent user (the demo presenter). Out-of-scope: multi-tenant load — the post-hackathon Supabase target adds RLS + connection pooling for that.	FastAPI uvicorn worker; numpy simulation runs in <50 ms; LLM is the bottleneck (≈15 s per cold analysis, instant on warm cache).
Reliability	`/api/analyze` must return a non-error response under every failure mode the LLM can produce — including unreachable, malformed JSON, schema-validation failures, and missing API key. ClawView (`/v1/clawview`) must degrade gracefully without blanking the page.	Each LLM stage has a deterministic fallback (`_mock_*` / `_heuristic_*`). The pipeline never returns 500 from an LLM problem. ClawView is wrapped in a class `ErrorBoundary` . Per PRD §9.2: 3 attempts / 120s read timeout default; Annotate tightens to 2 / 30s.
Maintainability	Single LLM entry point; schemas split by domain; endpoint handlers separate from services. Adding a fifth LLM call should require touching one service file and one schema file.	`core/glm_client.py` is the only place that talks HTTP to the model. `app/api/` holds handlers; `app/services/` holds orchestration; `app/schemas/` holds Pydantic contracts split into `common` , `policy` , `analyze` , `clawview` , `futureclaw` , `legacy_ai` packages.
Token / Latency	`/api/analyze` ≤ 15 s end-to-end (4 LLM calls, ~14 s LLM time + ~1 s overhead). ClawView ≤ 30 s worst case before degrading to mock. Slider drag stays at 60 FPS.	Streaming SSE on every LLM call; demo cache makes warm reruns instant; numpy keeps the LLM out of the slider loop; 5 s timeout buffer on async fetches with user-facing loading state.
Cost Efficiency	Average per-session token budget bounded; repeat queries on the same input never re-bill.	SHA-256 read-through demo cache de-duplicates identical reruns. Prompts are length-capped via `rag.py` lexical retrieval before the Extract call. The four-call pipeline is the upper bound; FutureClaw narrative regenerates only on mode toggle, never on slider drag.
Accessibility	WCAG 2.1 AA target.	Color-blind-safe risk palette (red ≠ green-only signal — shape + label too); keyboard navigation; ARIA labels.
Privacy	No PII in logs. PDFs not persisted in MVP.	UUID `analysis_id` only in logs; PDF bytes held in process memory for the analysis call lifetime, then dropped. PDPA-compatible deletion endpoint is part of the Supabase ship target.

3.3 Out of Scope / Future Enhancements

Scoped out of UMHackathon 2026 submission, kept on the post-hackathon backlog:

a. **Real-time insurer API integration** — no public Malaysian insurer offers one. b. **Actual policy purchase / cancellation** — regulated activity outside the decision-support remit (P3 anti-principle). c.

Claims filing assistance — separate workflow, separate user journey. d. **SME / business insurance** — different policy structures, different cost data. e. **Native mobile applications** — responsive web only for MVP. f. **Voice interface** — too risky to ship in 24 hours; scaffold endpoint stays in mock mode. g. **Mandarin / Tamil / Hokkien support** — BM + EN only per PRD P5. h. **Takaful Shariah-deep features** — current MVP is generic-takaful-aware but not Shariah-board-validated. i. **OCR for scanned PDFs** — PyMuPDF text-native only; Tesseract integration deferred. j. **Production frontend deployment** — localhost demo for the hackathon; Vercel + Render are the post-hackathon targets. k. **Supabase persistence layer** — Auth, RLS, pgvector RAG, Realtime streaming, PDPA deletion endpoint — all flagged in PRD §8.2 / §9.3 as ship target, not MVP-gating.

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4. Monitor, Evaluation, Assumptions & Dependencies

4.1 Technical Evaluation

4.1.1 Grayscale Rollout & A/B Testing

The hackathon submission is a single-machine demo with one user — there is no production rollout to grayscale against. The closest analogue is the `/v1/ai/status` endpoint, which exposes whether the backend is in **mock** or **live** mode; the frontend toggles a small badge based on that flag. This gives the same observability surface a real grayscale would.

Post-hackathon plan: when the Supabase + Vercel deployment lands, a new feature (e.g. an OCR ingestion path for scanned PDFs) will roll out behind a feature flag to 5% of authenticated users, monitor error rate and verdict-consistency drift for 24 hours, and graduate to 100% if the metrics hold. A/B testing two prompt variants for the Recommend stage is a natural first experiment because verdict consistency is already instrumented (see PRD F7 / `test_verdict_consistency.py`).

4.1.2 Emergency Rollback & Golden Release

PolicyClaw's Golden Release strategy is **multi-layer fallback**, not a CI/CD redeploy primitive. Every LLM stage is wrapped in three layers:

1. **Demo cache (Golden)** — `backend/data/demo_cache/<stage>_<sha256>.json` is pre-warmed for the three sample policies before the demo. A wifi outage or upstream API failure surfaces the cached payload with a "Served from demo cache" badge in the UI. This is the equivalent of automatic redeploy-to-Golden in a server context.
2. **Heuristic fallback (Emergency Rollback)** — when the cache misses *and* the LLM is unreachable / returns malformed JSON / fails Pydantic validation, the pipeline routes into `_mock_*` / `_heuristic_*` deterministic functions. Confidence band drops to LOW, the UI nudges the user toward a human advisor (P4 calibrated confidence).
3. **Schema validation guard (last line)** — `_extract_json_from_content` tolerates code-fenced LLM responses; on validation failure, the wrapper catches and routes into the heuristic path rather than 500-ing.

CI/CD-side: GitHub Actions runs a smoke-import + pytest hard-gate on every push (see §4.2). A failed gate blocks merge; rollback is `git revert` on the offending commit since the demo deploys directly from `main`.

4.1.3 Priority Matrix

Defines what triggers each emergency response and what the system does.

Priority	Trigger condition	Action	Owner
P1 — Critical	LLM upstream returns 5xx or times out (>120 s default; >30 s on Annotate).	`post_glm_with_retry` exhausts attempts, raises; calling stage catches and routes to heuristic fallback with LOW confidence band. UI shows "Limited annotation available" or equivalent. Demo continues.	shell
P1 — Critical	`OPENAI_API_KEY` absent in `backend/.env`.	Pipeline detects missing key at startup and enters mock / heuristic mode for all four LLM calls. `/v1/ai/status` reports `mode: mock`. Demo continues end-to-end with cached + heuristic outputs.	shell
P2 — High	PyMuPDF fails to extract text (scanned-only PDF with no text layer).	`parse_pdf_chunks` returns `[]`; `clawview_service` surfaces a low-confidence "limited annotation" response rather than a 500. UI shows an inline message; rest of flow continues.	clawview
P2 — High	Verdict drift across reruns (F7 break).	Demo cache write-through enforces byte-identical reruns. `test_verdict_consistency.py` is a CI hard-gate when present; a failure blocks merge.	shell
P3 — Medium	Frontend chart export fails on a niche browser.	Action Summary PDF still works (jsPDF is the demo path); chart PNG export is a polish feature, surfaces a toast on failure.	futureclaw
P3 — Medium	Style debt (`ruff` , `npm run lint`) regressions.	Lint is informational in CI (`continue-on-error: true`); merges proceed. Promote to hard gate post-hackathon.	shell

4.2 Monitoring

The MVP is single-machine and demo-time, so monitoring is minimal but explicit:

- **CI gate.** `.github/workflows/ci.yml` runs on every push and PR. Backend job: `pip install -r backend/requirements.txt` → smoke import `python -c "import app.main"` (hard gate) → `ruff check backend/` (informational) → `pytest backend/tests/ -q` (hard gate when tests present). Frontend job: `npm ci --prefix frontend` → `npm run lint --prefix frontend` (informational) → `npm run build --prefix frontend` (hard gate). 19 backend pytests are currently green.
- **Live status endpoint.** `GET /v1/ai/status` returns the live-vs-mock state and the configured model id. The frontend reads this once on session start and shows a badge if mock mode is active — judges see the truth at a glance.
- **Frontend error boundary.** `components/ErrorBoundary.tsx` wraps the ClawView and FutureClaw subtrees. A runtime exception in either feature surfaces a friendly inline message instead of blanking the page.
- **Demo cache observability.** Cache hits set `cached: true` on `AnalyzeResponse`; the frontend shows a "Served from demo cache" badge so the demo presenter knows the path being exercised.

Post-hackathon plan: Supabase Realtime + a lightweight health dashboard (Sentry / Logtail) cover error rate, p95 latency, and verdict-consistency drift.

4.3 Assumptions

State the operational and environmental conditions that must hold for the MVP to function as specified.

- **Text-native PDFs.** Sample policies are exported from insurer portals with a real text layer. Scanned-only PDFs are an explicit P2 fallback (limited annotation), not a primary path.
- **Stable internet for the live demo.** The demo cache is the contingency: if wifi drops, pre-warmed cached payloads serve every stage. The presenter can disable wifi pre-demo and the flow still works.
- **`OPENAI_API_KEY` available in `backend/.env`.** When unset, the pipeline enters mock / heuristic mode automatically — useful for offline iteration; the live-LLM path is the demo path.
- **Single concurrent user.** The demo presenter. Multi-user concurrency is post-hackathon.
- **Modern browser (Chrome 120+, Safari 17+, Firefox 121+, Edge 120+).** react-pdf-viewer + SVG overlay assume current browser engines; the experience is degraded on older browsers.
- **Malaysian context.** Cost data comes from BNM / LIAM / PIAM / MTA; outputs assume MYR. Non-Malaysian policies fall outside the demo scope.
- **English-or-Bahasa-Malaysia user.** Other Malaysian languages are out of scope per PRD P5.

4.4 External Dependencies

External tools and data sources the system relies on, each with a risk and a mitigation.

Tool / data	Purpose	Risk	Mitigation
OpenAI `gpt-5-mini` API	The four-role reasoning engine — Extract / Annotate / Score / Recommend, plus FutureClaw life-event narratives.	High — rate limits or upstream outages would gut every AI claim.	Streaming `post_glm_with_retry` with exponential-backoff retries. Demo cache pre-warmed for sample policies. Every stage has a deterministic mock / heuristic fallback. ClawView Annotate tightens to 2 / 30s so a slow upstream degrades within ~60s.
PyMuPDF (fitz) 1.26+	Text extraction with per-clause bounding boxes — required for the ClawView SVG overlay.	Medium — pip install can fail on uncommon platforms; no bbox = no ClawView wow factor.	Pinned in `requirements.txt`. `pdf_parser.parse_pdf_chunks` returns `[]` on garbage bytes; ClawView falls back to a side-pane risk list per PRD §10 critical-cut points.
BNM / LIAM / PIAM / MTA cost corpus	Real Malaysian medical-inflation series + life-event cost medians for FutureClaw.	Low — static JSON shipped under `backend/data/bnm_corpus/`, version-pinned.	Curated pre-hackathon; checksummed at load; cited with source URL on each scenario card.
react-pdf-viewer + jsPDF	Inline PDF rendering and client-side Action Summary export.	Low — both are stable npm packages with permissive licenses.	Pinned versions; jsPDF synchronous render path keeps Action Summary <2 s with no backend dependency.
GitHub Actions	CI gate on every push.	Low — outages affect velocity but not the demo.	Local-machine fallback: `pytest backend/tests/ -q` + `npm run build --prefix frontend` reproduce the gate.

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5. Project Management & Team Contributions

5.1 Project Timeline

The build window is 24 hours; the timeline maps to PRD §10. Each phase has a checkpoint.

Hours	Phase	Deliverables
0–2	Foundations	Next.js + FastAPI scaffolds running. LLM client smoke-tested. Three sample PDFs anonymized. Supabase setup attempted; deferred if it took >30 min — fell back to in-memory + cache as PRD §8.2 sanctions.
2–6	Backend core	`/upload` + PyMuPDF text extraction with bounding boxes. **LLM Call 1 (Extract)** working through streaming `post_glm_with_retry` + Pydantic validation. Schemas defined; BNM corpus seeded.
6–10	ClawView (Wow 1)	**LLM Call 2 (Annotate)** returns per-clause risk levels. react-pdf-viewer + SVG bounding-box overlay shipped. Click-to-tooltip working. *Checkpoint: ClawView visibly working on one policy.*
10–14	Health Score + Recommendation	**LLM Call 3 (Score)** → four sub-scores → circular gauge. **LLM Call 4 (Recommend)** → verdict + reasons + confidence → VerdictCard.
14–18	FutureClaw (Wow 2)	numpy Monte Carlo working. Affordability chart + sliders. Life Event mode with four scenarios (Cancer / Heart Attack / Disability / Death). LLM narrative pass. *Checkpoint: both wow factors demonstrable.*
18–21	Polish	Action Summary jsPDF download. BM/EN toggle. Framer Motion transitions, loading states, error boundaries.
21–23	Testing & docs	19 pytest units green (extraction, simulation, FutureClaw, verdict consistency). README + architecture section. PRD / SAD / QATD final pass.
23–24	Demo prep	Three full demo run-throughs. Four-minute screen capture as backup. Demo cache pre-warmed for the three sample PDFs.

Critical cut points. If at Hour 12 the core flow isn't end-to-end working: cut F9 (EN only), cut Life Event mode (Affordability only), cut PDF download (screen-only summary). If at Hour 18 ClawView isn't aligned: fall back to a side-pane risk list with page references — loses the *wow* but ships.

5.2 Team Members & Roles

Member	Role	Responsibilities
Solo developer	Full-stack engineer	Backend FastAPI services, frontend Next.js components, LLM prompt engineering, Monte Carlo simulation, CI configuration, the SAD / PRD / QATD documentation set.
Claude Code (paired AI)	Pair-programming assistant	Schema scaffolding, refactoring passes, test authoring, documentation drafting. Reviewed by the human developer before commit.

This is a solo-dev hackathon entry per `CLAUDE.md`'s framing; there is no team to split roles across. The role row is included to align with the SAD template — readers should treat all of "Backend Lead", "Frontend Lead", "QA Lead", "DevOps", "Tech Writer", and "Product Owner" as collapsed onto the solo developer.

5.3 Recommendations

Forward-looking improvements after the hackathon submission, ordered by expected impact.

1. **Wire Supabase as the persistence + auth layer.** Per PRD §8.2 the Supabase target unlocks pgvector RAG (real semantic retrieval over the BNM corpus instead of `rag.py`'s lexical baseline), magic-link Auth, RLS-enforced PDPA deletion, and Realtime streaming so multi-stage agents can broadcast progress to a future demo. Setup is ~15–20 minutes once outside the 24-hour window.
2. **Promote `ruff` and `npm run lint` to hard CI gates.** Currently informational so style debt cannot block the 24-hour critical path. Once stable, these should fail the build.
3. **Real OCR ingestion for scanned PDFs.** Tesseract or a hosted equivalent expands the addressable user base — a meaningful share of Malaysian policies still arrive as image-only PDFs.
4. **Embedding-based retrieval in `rag.py`.** Replace the lexical scorer with `text-embedding-3-small` (or Supabase pgvector) once the corpus grows beyond what fits comfortably in context.
5. **Frontend Vitest + Playwright suite.** The 24-hour budget excluded both; manual QA + the CI build gate carry the frontend today. Add unit tests for the `t()` helper and the BM/EN toggle, plus a Playwright golden-path E2E.
6. **Redis or Vercel KV for the cache layer.** The on-disk SHA-256 cache is fine for one-machine demo; multi-instance deployments need a shared cache to keep verdict consistency across replicas.
7. **Real BM-fluent localization review.** The current BM strings come from the LLM; a native-speaker copy edit will sharpen them before any public release.

End of SAD. Paired with `PRD.md` v2.2 (product spec) and `QATD.md` v1.0 (risk + test plan). Decision-support software, not licensed financial advice.